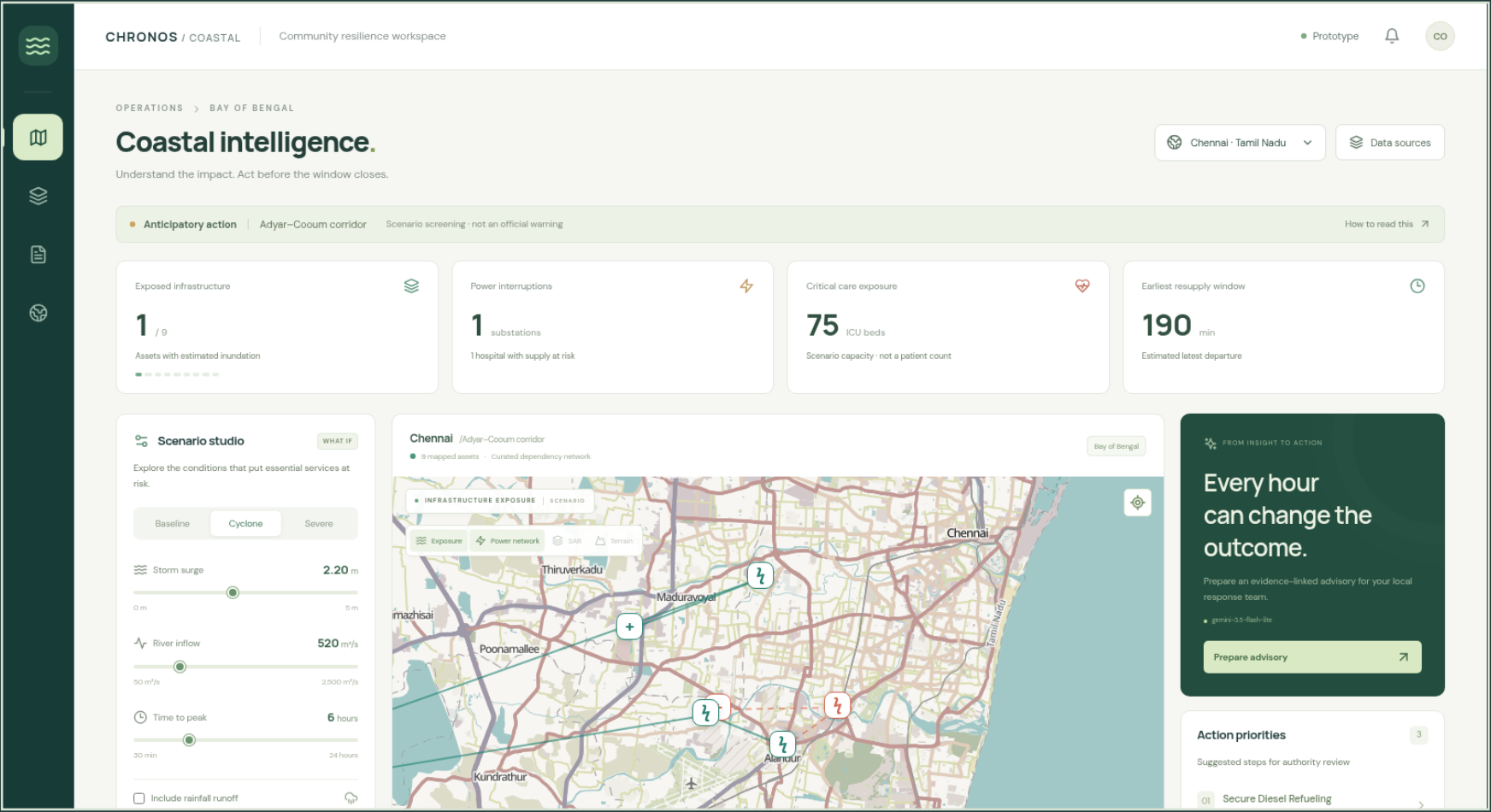


A dry hospital can still lose power.

See the dependencies.
Act before access closes.

A coastal preparedness workspace connecting flood scenarios to electricity, critical care, and resupply access.

4 corridors • connected service impacts



A working demonstration with explicit model assumptions.

Flood exposure is only one part of the risk.

Essential services depend on assets beyond their own footprint.

01

Compound water

Surge and river inflow interact in the screening model.

02

Power dependency

A low-lying upstream substation can interrupt supply.

03

Dry hospital

A facility can remain dry while its grid connection fails.

04

Backup reserve

Generator fuel becomes a service-continuity dependency.

05

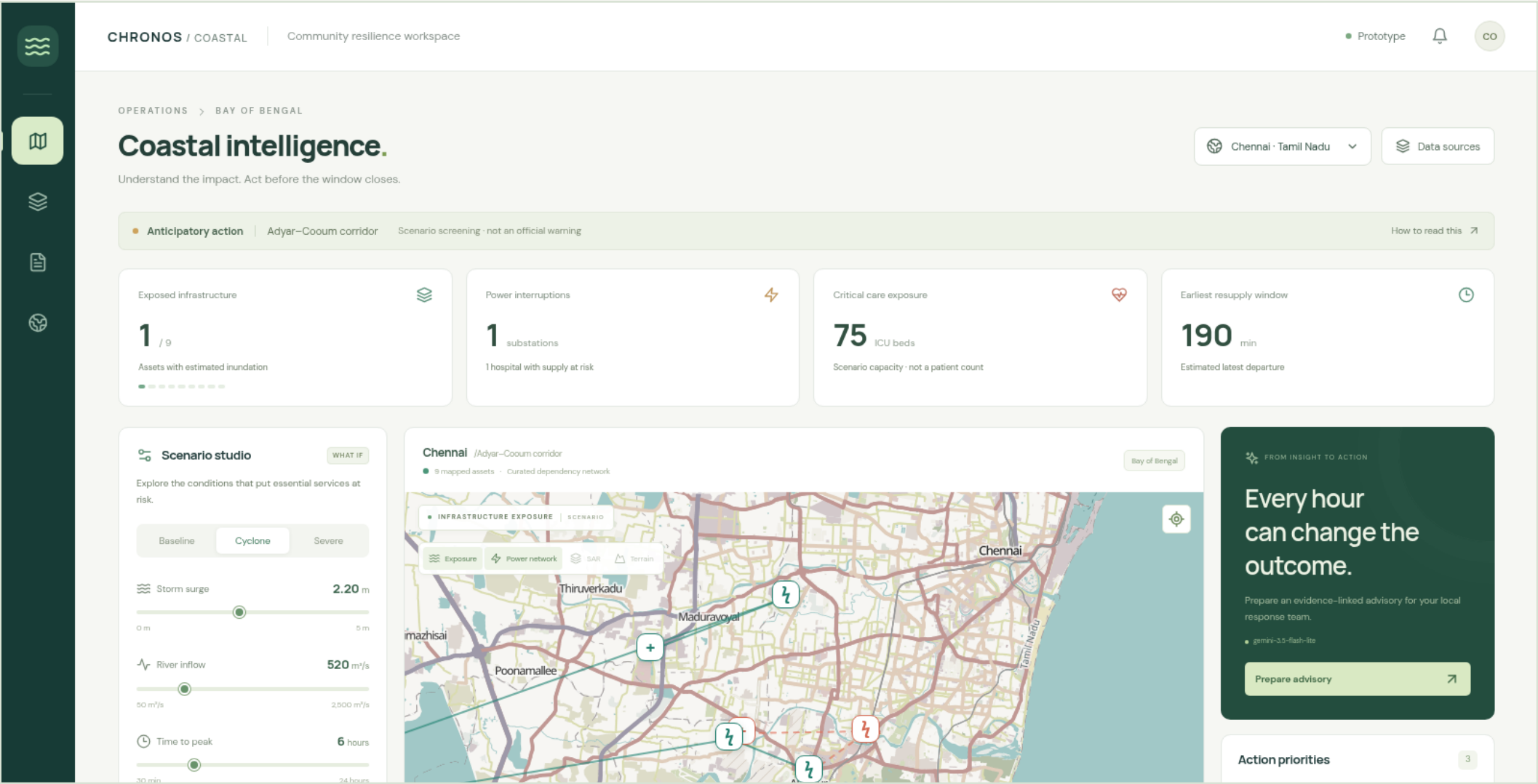
Resupply access

Choke-point depth and travel time limit delivery windows.

Our question: which service fails next, and what response window remains?

From “what if” to a reviewable action.

One workspace connects scenario controls, service impacts, and evidence-linked advisories.



Explore

Choose a corridor. Compare baseline, cyclone, severe, or custom inputs.

Trace

Inspect power links, hospital backup reserves, and resupply windows.

Prepare

Generate a draft, download a brief, and verify test delivery.

Dry ground. Interrupted supply.

Selected scenario: 1.85 m surge • 550 m³/s river inflow • 6 h to peak

VPS Lakeshore Multi-Speciality Hospital

0.00 m Estimated depth at hospital

53.3 h Modeled generator reserve

68 Demonstration ICU-bed capacity

Demonstration assumptions; this is not a live facility-status report.

INFRASTRUCTURE DETAIL

VPS Lakeshore Multi-Speciality Hospital

Estimated depth

0.00_m

Ground elevation

3_m

Asset ID

HI_Lakeshore

Coordinates

9.9191, 76.3191

Category

healthcare

Grid supply

Interrupted in scenario

Fuel reserve

53.3 hours

Critical care capacity

68 ICU beds

This hospital is dry in the scenario, but an upstream power dependency is interrupted. Check backup generation and resupply access.

Elevation, equipment thresholds and network links are curated demonstration assumptions.

Locate on map

Protect the resupply window.

Departure deadline = modeled clearance breach time – route travel allowance.

Medical oxygen

175 min

Estimated latest departure from run start

Kundannoor Bridge Approach (NH-66 Bypass)

Assumed clearance: 0.20 m

Transit allowance: 25 min

Diesel fuel

246 min

Estimated latest departure from run start

Container Terminal Road Bridge Abutment

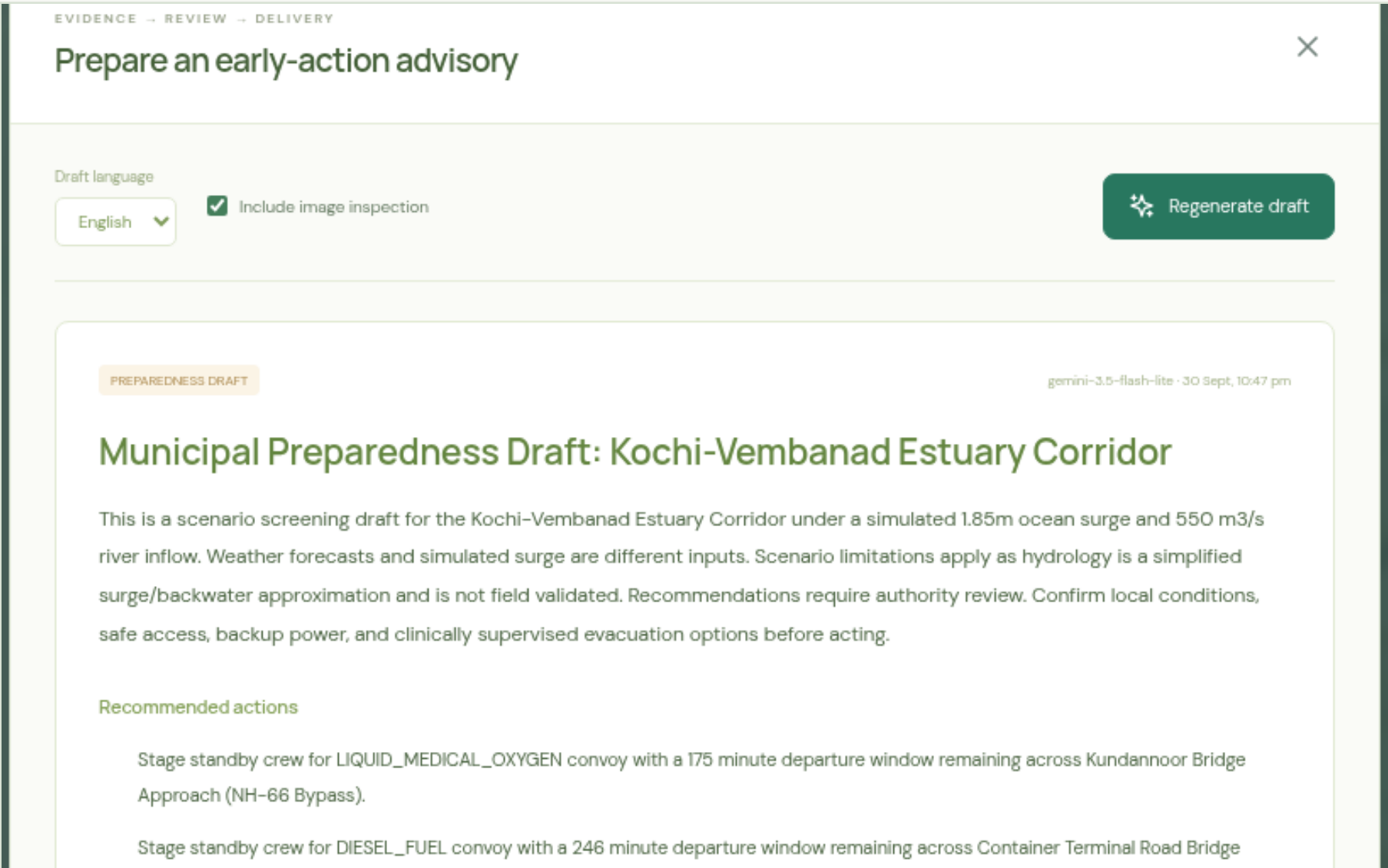
Assumed clearance: 0.45 m

Transit allowance: 22 min

These are screening estimates, not live navigation or dispatch authorization.

Gemini turns scenario evidence into action.

Live Google Gemini drafts a structured advisory; the exported brief records the actual model mode.



Live Google AI

Gemini 3.5 Flash Lite reasons over scenario evidence and returns recommendations for review.

Immutable evidence

Checksummed drafts and delivery receipts persist independently. Repeated requests share one receipt.

Reviewed test delivery

Download the brief. Deliver to the in-app inbox. Another browser session has a separate inbox.

A public prototype. A reproducible backend.

The repository includes the full Python server and the hosted runtime adapter.

Browser workspace

React + Vite + Leaflet

Responsive dashboard

Scenario sharing and JSON export

Advisory review and brief download

Hosted adapter

Public Sites deployment

D1 records scoped by session

Hydro / grid / logistics parity

Test inbox only on the public demo

Python backend

FastAPI + NetworkX + NumPy

SQLite persistence + Docker

Diagnostic APIs + event streaming

Optional authenticated Earth Engine

Providers stay explicit

Live Google Gemini drafting · Open-Meteo forecast context · satellite observations require configuration

Tested behavior. Honest scope.

Engineering checks establish repeatable prototype behavior, not field accuracy.

58

Backend tests passed

6

Hosted API tests passed

100

Cross-runtime parity cases

4

Sample coastal corridors

Also verified

Public desktop and mobile flow; downloads and receipts; session isolation; provider-outage behavior; Docker build and container runtime in GitHub Actions.

Next: BRICS partner pilots

Adapt for Brazil, China, Russia, and South Africa with licensed local asset data. Calibrate hydraulics and validate languages with local partners.

No independent field validation, official warnings, confirmed medical status, or financial transfers are claimed.

EXPLORE THE WORKING PROTOTYPE

Earlier action starts with clarity.

Built for communities.

Designed for connected preparedness.

Public prototype

chronos-coastal.onrender.com/

Repository + submission release

github.com/j33v4nz/chronos-coastal



SCAN TO EXPLORE